

Evaluating the Impact of New Ethanol Plants on School District Finances and Student Outcomes

Pierce Plucker

Project Notes

This project was supported by South Central Sun Grant Summer Fellowship Program (2025), and this document is a slightly-modified report I submitted when the program completed. Due to the regional scope of the grant program, the states analyzed are Missouri and Kansas due to the regional scope of the program, but this will be expanded to states with more plants, such as those in the Midwest and Great Plains. Other future directions for this project are discussed in the final section.

Project Summary

This project examined the economic and educational impacts of bio-ethanol plants in Kansas and Missouri, with a focus on public K-12 school district finances. A key contribution of this project was the construction of a novel, longitudinal data set containing district finances, district boundaries, plant locations, and plant opening dates which may prove useful in answering research questions outside of this project. Leveraging these data and modern statistical techniques, the empirical analysis investigates the impacts of new ethanol plant openings via the staggered-adoption synthetic controls method (SA-SCM) of Ben-Michael et al. (2022). Six district-finance outcome variables—trimmed down from a larger set due to data availability and findings detailed in the next section—were analyzed: enrollment, total revenue per pupil, local revenue per pupil, local revenue as a share of total revenue, total expenditure per pupil, and instruction expenditure per pupil. Of these, plant openings only had a significant (90% CI) effect on local revenue at 7 years post-opening. While not staggering in terms of statistical significance, the results serve to guide future research endeavors regarding the local economic impacts of bio-ethanol plants. Additionally, ethanol plants were not found to have an impact on dropout rates in Kansas.

Data Construction and Methodology

EDGE school district boundaries and district-level finance data were gathered from National Center for Education Statistics (2024b,a), taking all school districts in Kansas or Missouri from 1992 through 2022. The time dimension for school finance data is the academic year recorded such that AY 2010-2011 corresponds to 2011. Observations before 1995 are dropped from the data set because although finance data for AY 1990 and 1992 are publicly available, *annual* NCES finance data collection began in 1995. Annual dropout rates were available for all years from Kansas State Department of Education (n.d.). Dropout rate data for Missouri are sparse for many districts: none of the Missouri districts with an ethanol plant have complete dropout series. Moreover, the few districts with availability from 2000 onward are high-enrollment districts which are unlikely to serve as good controls for districts with plants.

The district boundaries used for this project are those recorded in 2022. Consolidations, closings, and mergers were manually accounted for based on reported district boundary changes (National Center for Education Statistics, 2024b). Ethanol plants and production capacity data from U.S. Energy Information Administration (2024) are merged with their respective geographic coordinates from U.S. Environmental Protection Agency (2024). This data set was then merged with the EDGE, finance, and KS dropout data sets. The date of opening for each plant was manually collected from each plant’s website if available.¹ The date of opening is recorded in the data as the calendar year of first ethanol production. Matching finance data academic years and calendar years for plant openings deserves further attention as this project continues beyond this summer.

The SA-SCM methodology of Ben-Michael et al. (2022) builds on traditional SCM methods, allowing for multiple treated units and staggered treatment adoption. In the present context, a treated unit is a district which had a new ethanol plant opening in the years 2001-2022. As with any method in the SCM family, synthetic counterfactuals are constructed by weighting donor units (districts which have never had a plant) to closely match the pre-treatment trends of treated units. The central innovation of SA-SCM is the use of *partially-pooled weights* which effectively interpolates between separately fitting SCM weights for each treated unit and fitting a single SC for an average treated unit. The result is a reduction in bias that may arise had the researcher exclusively chosen one of the two methods. More pre-treatment periods of data lead to a better pre-treatment fit. Because of this, any plant that opened before 2002 was stripped of its treated status and dropped from the donor pool. Inference on post-treatment effects also relies on the number of post-treatment periods. As such, districts with ethanol plants which opened after 2018 were also stripped of their treatment status and removed from the donor pool.

Figure (1) displays the school districts in Kansas and Missouri. Gray districts are those that are either not K-12 (PK-12) or not fully-public school districts. Missouri has some districts that are designated as both public and charter. Gray districts are therefore not included in the donor pool because all of the treated units are public-only and K-12 (or PK-12) districts, and the funding structure may differ for elementary-only districts or charter districts. Blue districts comprise the donor pool, and are weighted to form synthetic green (treated) districts in the analysis. Brown districts are those involved in a merger/consolidation between 1995 and 2022, yellow are those with a plant opening before 2002 or after 2017, and purple are those which have a plant that has an annual production capacity less than 20 million gallons per year.

Objectives

Objectives 1 and 2: Tax Base and Finance Structure Analysis

Figure (2) contains the SA-SCM results with the dark line representing the average effect of a plant opening on district finance variables. The shaded area represents 90% confidence intervals associated with estimates for each year post-plant opening.² Total revenue per pupil remains largely unchanged in the years following a plant’s first ethanol production. This is the expected result due to district funding (equalization) formulas determined at the state level. Neither total expenditure or expenditure per pupil respond significantly to new ethanol plants, which is also unsurprising due to funding-formula constraints. However, local revenue as a proportion of total revenue increases

¹The website for Golden Triange Energy LLC in Craig, MO does not list the date of first ethanol production, and further efforts to track down this information were unfruitful. Therefore, the year of this plant’s opening was recorded as zero and appears as the small yellow district in northwest MO in Figure (1).

²All estimations use only the lagged outcome for balancing pre-treatment trends.

by roughly 7% in the seventh post-treatment year. Is this increase driven by the added taxable base of the ethanol plant itself (primary effect), or by a boost in economic activity related to the plant (secondary effects)? This question remains to be answered concretely, and is further muddled by tax incentives offered to ethanol producers. If all plants were given a 100% abatement agreement (at least spanning the 8-year post-opening period), then the boost in local revenue share could be more clearly attributed to secondary effects of the plant.

Objective 3: Student Outcomes

Data constraints limited the analysis to dropout rates in Kansas school districts. Results in Figure (3) suggest ethanol plants did not have an impact on dropout rates. In contrast to Rickman et al. (2017), who find that shale booms reduced high school attainment, the comparatively smaller scale and labor demand of ethanol plants likely generates too modest of a shock to alter student schooling decisions.

Future Directions

This project can be extended by incorporating further, potentially restricted-access data, such as incorporating funding formulas explicitly in the empirical analysis, obtaining abatement agreement data, expanding the sample to states with more plants (Great Plains). Further analysis of the secondary effects of ethanol plants is also a direction of interest including potential basis-driven land-value boosts. Incorporating additional measures of student achievement would further strengthen our understanding of how ethanol plant development affects educational outcomes.

An additional avenue of interest is cross-state analysis. Each state has a different funding formula which may lead to different effects in each state depending on degrees of redistribution. We may also look into using plant shutdowns as a potential source of exploitable variation. A final thought is to use generic machine learning to uncover heterogeneous patterns.

References

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Figure (1) School Districts of KS and MO

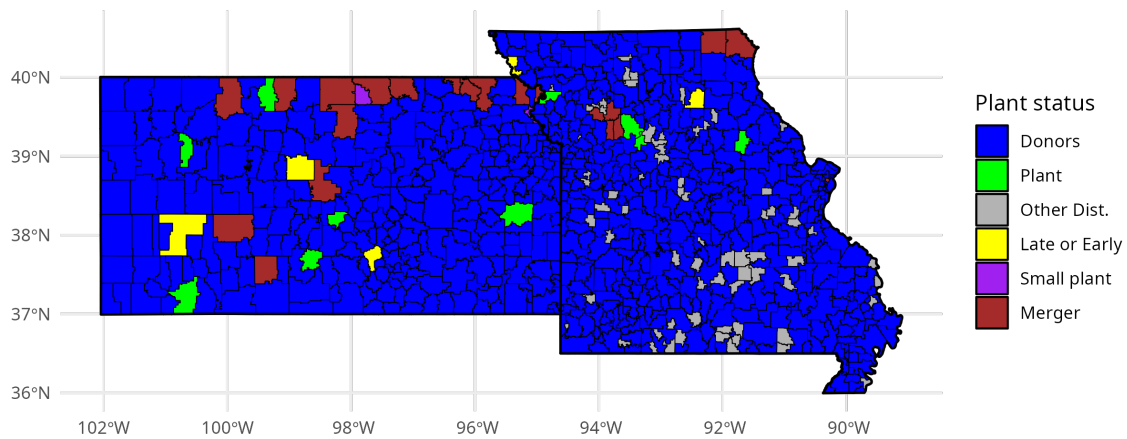


Figure (2) Impacts on District Finance Outcomes

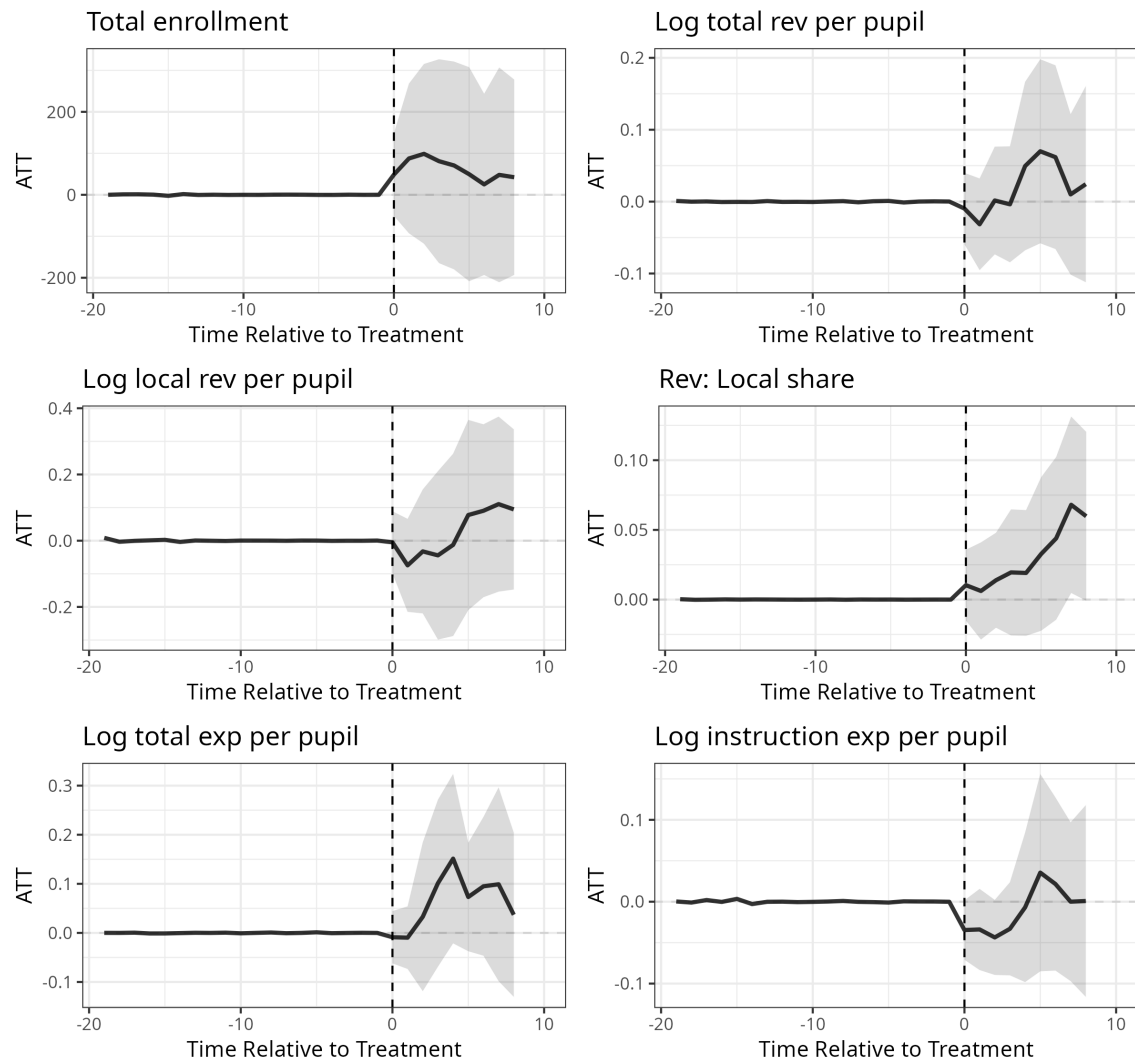


Figure (3) Dropout Rates for Kansas School Districts

